

Introduction to Data Visualization

Dr. Anilkumar V. Brahmane | Computer Engineering | Sanjivani College of Engineering

Data visualization is the graphical representation of data to reveal patterns, trends, and insights that are not easily seen in raw numbers. We use charts, graphs, and maps to communicate complex information in a simple, understandable way.

Charts & Graphs

Turn numbers into pictures

Find Patterns

Spot trends at a glance

Make Decisions

Evidence-based insights

Python Tools

Matplotlib, Seaborn, Plotly



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Introduction to Matplotlib

Matplotlib is a **comprehensive and widely-used data visualization library in Python**. It provides a flexible platform for creating static, animated, and interactive visualizations. It can plot a wide variety of chart types to suit any data analysis need.



Histogram

Shows frequency distribution of data



Pie / Donut Chart

Shows proportions of a whole



Box Plot

Shows data spread and outliers



Scatter Plot

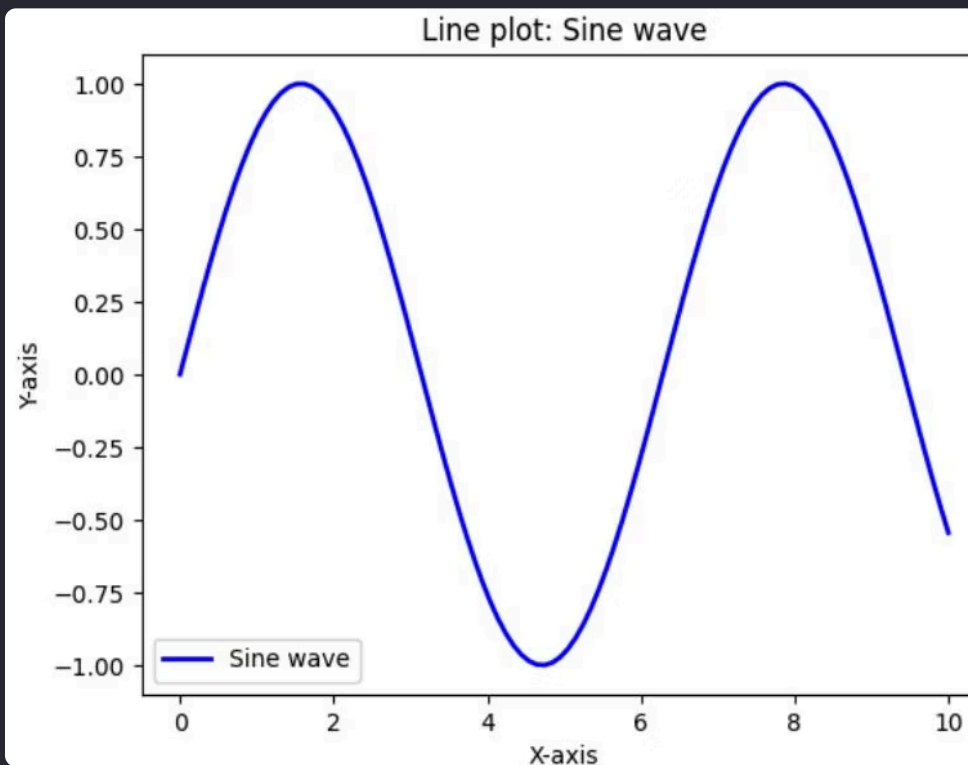
Shows relationship between variables

Line Plot & Scatter Plot

Line Plot

Displays data points connected by straight lines. Best for showing **trends over time or a continuous interval**.

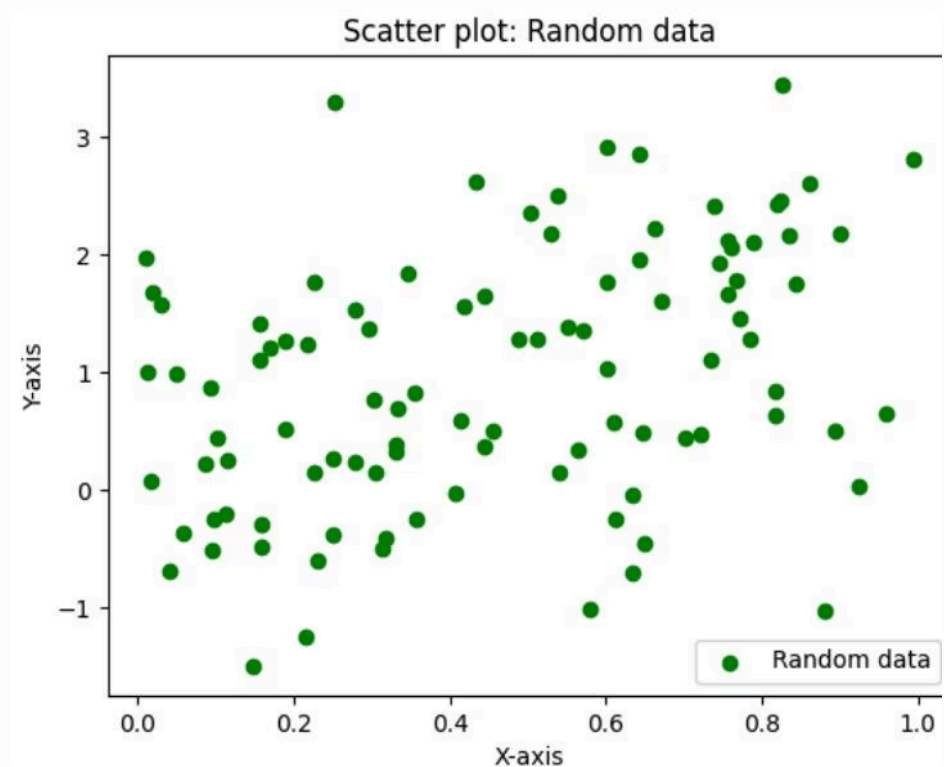
```
x = np.linspace(0, 10, 100)
y = np.sin(x)
plt.plot(x, y, label='Sine wave',
         color='blue', linewidth=2)
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Line plot: Sine wave')
plt.legend()
plt.show()
```



Scatter Plot

Displays the **relationship between two continuous variables**. Each point represents one observation. Useful for finding patterns, clusters, or trends.

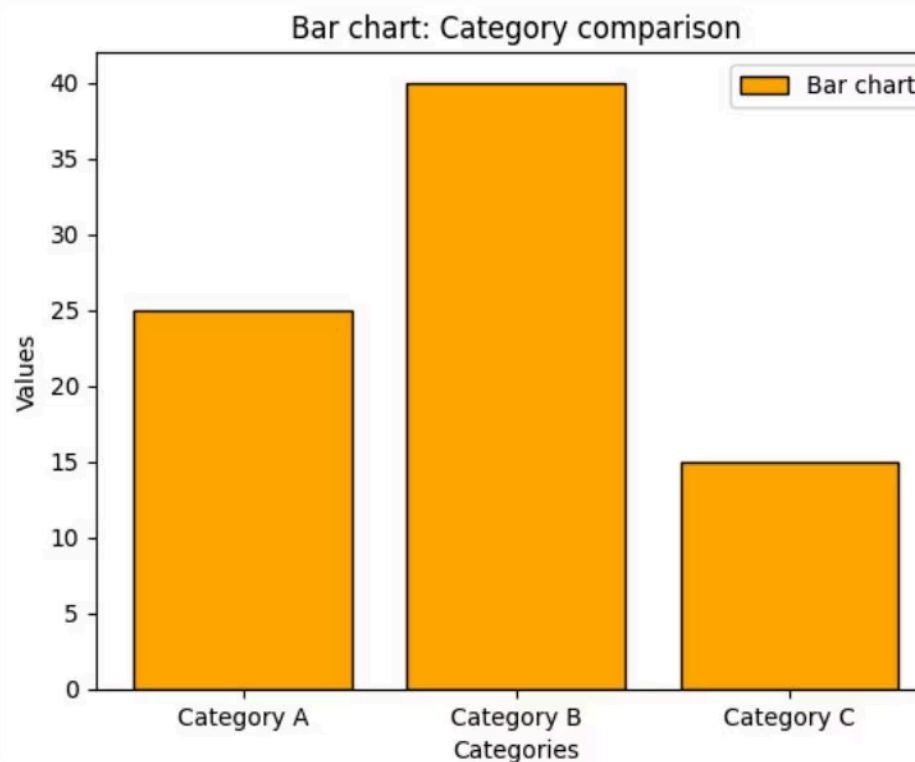
```
x = np.random.rand(100)
y = 2 * x + np.random.randn(100)
plt.scatter(x, y, color='green',
            marker='o', label='Random data')
plt.xlabel('X-axis')
plt.ylabel('Y-axis')
plt.title('Scatter plot: Random data')
plt.legend()
plt.show()
```



Bar Chart & Box Plot

Bar Chart

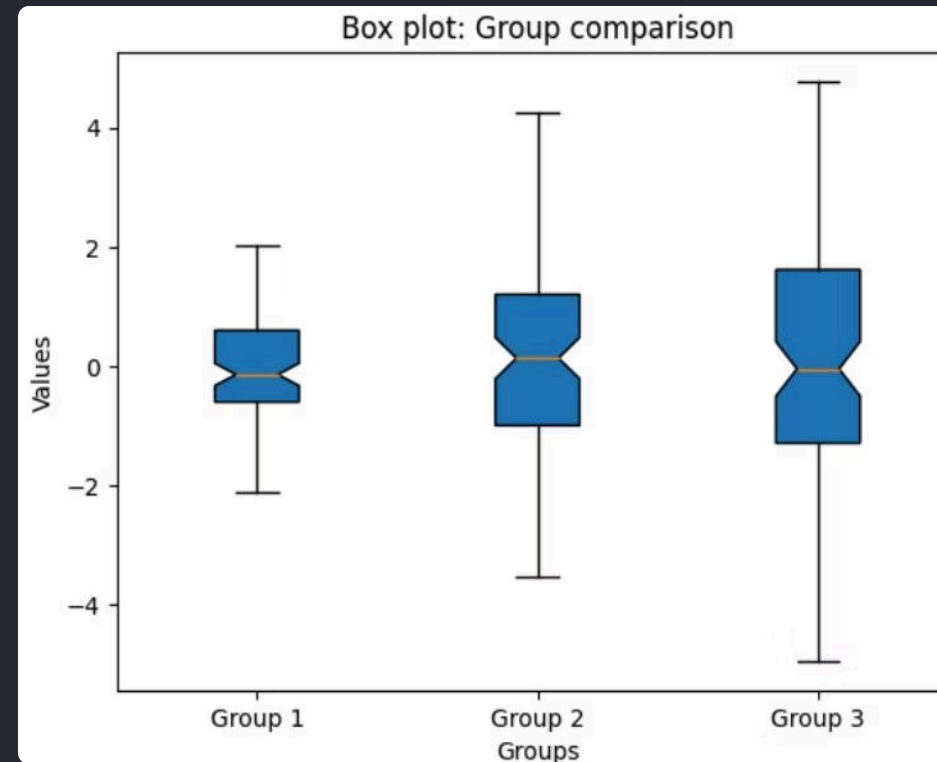
Uses **rectangular bars** to compare sizes or frequencies of different categories. Bar length is proportional to the value it represents.



```
categories = ['Cat A', 'Cat B', 'Cat C']
values = [25, 40, 15]
plt.bar(categories, values,
        color='orange', edgecolor='black')
plt.title('Bar chart: Category comparison')
plt.show()
```

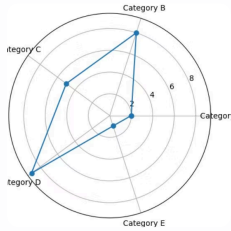
Box Plot

Shows the **distribution of a dataset** — minimum, Q1, median, Q3, and maximum. Great for spotting outliers and understanding data spread.



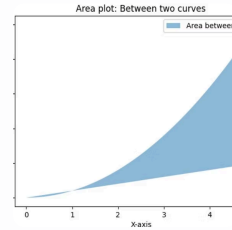
```
data = [np.random.normal(0,1,100),
        np.random.normal(0,1.5,100),
        np.random.normal(0,2,100)]
plt.boxplot(data,
            labels=['G1','G2','G3'],
            patch_artist=True, notch=True)
plt.show()
```

Radar, Area, Polar, Treemap & Pie Chart



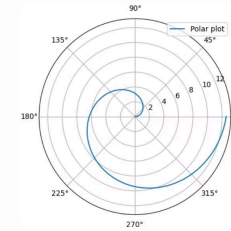
 Radar Chart

Displays **multivariate data** on axes from the same point. Good for comparing multiple categories at once.



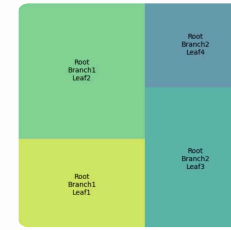
 Area Plot

Shows **cumulative quantities** over a continuous interval. Illustrates each variable's contribution to the total.



 Polar Plot

Represents data in **polar coordinates**. Useful for visualizing cyclic patterns and relationships.



 Treemap

Visualizes **hierarchical data** as nested rectangles. Easily compare sizes of branches and identify patterns.

 **Pie Chart:** A circular chart divided into slices — each slice shows a proportionate part of the whole (total = 100%). Used to visualize distribution of a categorical variable. Use `plt.pie(values, labels=labels, autopct='%1.1f%%')` in Matplotlib.